Yuekai Kevin Lu

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Education

Tufts University *Medford, MA*
B.S. in Mechanical Engineering; Minor in Economics *Expected May 2027*
GPA: 3.77; Dean's list all semesters

Experience

Eaton Corporation (Power Distribution Monitoring and Control) *Asheville, NC*
Mechanical Design Intern *May 2026 – August 2026*

- Gathered Voice of Customer (VOC) feedback and translated Critical to Quality (CTQ) requirements into engineering specifications, defining project scope for a variable transformer (Variac) enclosure and model low-voltage switchgear (LVS) tradeshow display
- Developed hand-sketched concepts and created over 300 detailed 3D CAD models in Autodesk Inventor for the Variac enclosure and model LVS, incorporating feedback from two formal design reviews to improve functionality, manufacturability, and usability
- Produced complete engineering documentation including electrical wiring schematics, bills of materials (BOMs), and assembly documentation to support prototype fabrication and system integration
- Managed prototype procurement by generating custom in-house fabrication orders and purchase requests for outsourced components, coordinating manufacturing and sourcing activities to meet project deadlines
- Assembled and validated the Variac enclosure prototype, verifying component fit and assembly while advancing the model LVS project through completion of the CAD design phase

Eaton Aerospace Corporation (Fluid and Electrical Distribution Division) *Beltsville, MD*
Quality Engineering Intern *May 2025 – August 2025*

- Conducted time studies to analyze inspection workflows, identify bottlenecks, and propose efficiency improvements
- Authored and revised 115 inspection plans in Net-Inspect, ensuring alignment with quality standards, customer requirements, and process capability (CpK) metrics
- Trained 7 inspectors on digital inspection and data collection processes to lead a division-wide transition to statistical process control (SPC)
- Created standard work instructions for both inspection and shipping teams, including procedures for reviewing FAIRs and inspection reports
- Designed a cost analysis spreadsheet to quantify process improvements and directly link quality initiatives to business objectives

Eaton Aerospace Corporation (Fuel and Motion Control Systems Division) *Grand Rapids, MI*
Manufacturing Engineering Intern *May 2024 – August 2024*

- Collaborated with a team of two engineers in designing and implementing unique labels for certain sub-system assemblies, ensuring precise identification for customers and seamless integration in production
- Modeled 3D fixtures and technical drawings utilizing Creo CAD software to accurately replicate and improve existing equipment
- Authored detailed work instructions for the testing process of multiple units, supporting consistency and accuracy in quality assurance
- Designed and fabricated transport vehicles to distribute parts across the plant, boosting site productivity and operational flow

Leadership & Campus Roles

Tufts University *Medford, MA*
NOLOP Makerspace Student Staff *August 2026 – Present*

- Operate and maintain 3D printers and laser cutters while training students on equipment and shop safety

Mechanical Engineering Department Peer Mentor *August 2026 – Present*

- Mentor underclassmen on coursework, course selection, and career paths in mechanical engineering

Skills

Technical: SOLIDWORKS, Creo, COMSOL, Microsoft Office Suite, MATLAB, C++, Autodesk Inventor
Fabrication: 3D printing, water-jet operation, MIG welding, laser cutting
Languages: Fluent English, Conversational Mandarin Chinese (Suzhou dialect)
Interests: Formula One, Powerlifting, American Football, Poker